UNIVERSAL MOTHER-WAVELET DILATION AND THE PHASE-INDEPENDENT CONFOCAL ELLIPSE IDENTITY IN THE SELBERG CLASS VERSUS BREAKDOWN OF PRIME-GAUGE PURITY AND CHLADNI CHAMBER TOPOLOGY IN THE DAVENPORT–HEILBRONN COUNTEREXAMPLE

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October 25, 2026

Abstract. We establish a dichotomy between universal harmonic laws for self-dual Dirichlet series with a Gamma functional equation and Euler-product closure mechanisms separating Selberg-class L-functions from the Davenport–Heilbronn counterexample $\mathcal{D}_{\mathrm{DH}}(s)$ (Titchmarsh 1986, §10.25). First, for any self-dual completed L-function $\xi_f(s) = Q_f^s \gamma_f(s) L(s,f) = \xi_f(1-s)$ of degree $d_f > 0$ (including $\zeta(s)$, $L(s,\chi_5)$, Ramanujan's cusp form $L(s,\Delta)$ of degree 2, and $\mathcal{D}_{\mathrm{DH}}(s)$), we prove the mother-wavelet decomposition $\Phi_f(u) = \sum_{m=1}^{\infty} a_f(m) m^{-1/2} \varphi_1^{(f)}(u + \frac{1}{2}\ln m)$, the branch-damping inequality $\rho_{\Gamma,f}(d,t)|a_f(n)|n^{-1/2+d} < \mathrm{e}^{-2d\omega_{n,Q_f}(t)}|a_f(n)|n^{-1/2-d}$, and the phase-independent confocal ellipse identity $|T_n^{(f)}(d,t)|^2 - |T_n^{(f)}(0,t)|^2 \equiv 4\frac{|a_n|^2}{n}\sinh^2(d\omega_{n,Q}(t)) \geq 0$. Second, whereas Euler-product L-functions satisfy prime-power purity $b_N^{\mathcal{E},(f)}(X) = \frac{\Lambda_f(N)}{\ln N}$ on $(X, X^2]$, $\mathcal{D}_{\mathrm{DH}}(s)$ suffers a multiplicativity defect $\delta_{\mathrm{DH}}(m,n) = \mp(1+\kappa^2) \approx \mp 1.080701$ at $N = 4, 6, 9$, supplying 54.03% composite destructive interference at $(d_*, \gamma_*) = (0.30851718, 85.69934849)$ and triggering a four-fold breakdown across Chladni walls, $\widetilde{W}_{\mathrm{DH}}$, Hadamard flux $\pi\,\Theta(d - d_*)$, and the de Branges Sylvester index $\nu_- : 0 \to 1$.

1. Introduction and Main Results

A foundational benchmark in analytic number theory, emphasized by Titchmarsh [31, §10.25] following the construction of Davenport and Heilbronn [5] (see also Selberg [27], Balanzario and Sánchez-Ortiz [1], and Bombieri and Ghosh [2]), is the contrast between automorphic L-functions in the Selberg class that possess an Euler product and linear combinations of Dirichlet series that satisfy the same functional equation without multiplicativity. Let χ_5 denote the primitive odd Dirichlet character modulo 5 defined by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$. The Davenport–Heilbronn function is defined by the real-coefficient Dirichlet series

$$\mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s,\chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s,\overline{\chi_5}) = \sum_{n=1}^{\infty} a_{\mathrm{DH}}(n)\, n^{-s}, \qquad \kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}, \tag{1.1}$$

where the 5-periodic coefficients are $a_{\mathrm{DH}}(1) = 1$, $a_{\mathrm{DH}}(2) = \kappa \approx 0.2840790438$, $a_{\mathrm{DH}}(3) = -\kappa$, $a_{\mathrm{DH}}(4) = -1$, and $a_{\mathrm{DH}}(5) = 0$. Because the Gauss sum of χ_5 satisfies $\tau(\chi_5) = \mathrm{i}\sqrt{5}$ and $\frac{1+\mathrm{i}\kappa}{1-\mathrm{i}\kappa} = \frac{\mathrm{i}\tau(\chi_5)}{\sqrt{5}}$

Date: October 25, 2026.

2020 *Mathematics Subject Classification.* Primary 11M41, 11M26; Secondary 11F66, 30D15, 46E22.

Key words and phrases. Selberg class, Davenport–Heilbronn function, automorphic L-functions, Ramanujan Δ cusp form, universal mother wavelet, confocal ellipse identity, exponential prime-gauge factor, Chladni nodal chambers, Hadamard–Krein flux jump, de Branges Sylvester inertia.

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is tuned to unity, the completed function

$$\xi_{\mathrm{DH}}(s) := \left(\frac{\pi}{5}\right)^{-(s+1)/2} \Gamma\!\left(\frac{s+1}{2}\right) \mathcal{D}_{\mathrm{DH}}(s) = \xi_{\mathrm{DH}}(1-s), \qquad \Xi_{\mathrm{DH}}(z) := \xi_{\mathrm{DH}}\!\left(\tfrac{1}{2} + \mathrm{i}z\right) \tag{1.2}$$

($z = t - \mathrm{i}d$, $s = \frac{1}{2} + d + \mathrm{i}t$) satisfies the exact same self-dual $\Gamma(\frac{s+1}{2})$ functional equation and even Poisson–Jacobi kernel symmetry $\Phi_{\mathrm{DH}}(u) = \Phi_{\mathrm{DH}}(-u)$ as an odd primitive L-function [26, 12, 25]. Yet because $a_{\mathrm{DH}}(mn) \neq a_{\mathrm{DH}}(m)a_{\mathrm{DH}}(n)$, $\mathcal{D}_{\mathrm{DH}}(s)$ has no Euler product and possesses infinitely many zeros off the critical line $d = 0$, beginning in the critical strip with the first off-line conjugate pair at $s_0 = \frac{1}{2} + d_* + \mathrm{i}\gamma_* \approx 0.80851718 + 85.69934849\mathrm{i}$ [30, 1].

Any analytical framework addressing the transverse Wronskian $W_f(d,t) := \frac{1}{2}\partial_d |\Xi_f(t - \mathrm{i}d)|^2 = R_f\,\partial_t I_f - I_f\,\partial_t R_f$ [29, 9, 13] and Wigner phase-space positivity [32, 10, 4] must therefore answer two precise structural questions:

(1) *Universal Archimedean and single-mode symplectic geometry.* Which identities of the logarithmic u-domain representation $\Xi_f(z) = 4\int_{-\infty}^{\infty} \Phi_f(u)\,\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u$ and of the individual combined forward-plus-reflected Riemann–Siegel modes $T_n^{(f)}(d,t)$ [28, 8] hold universally for *every* self-dual Dirichlet series of degree $d_f > 0$, regardless of whether the coefficients $a_f(n)$ are multiplicative?

(2) *Exact locus of non-multiplicative breakdown.* When we apply the exponential prime-gauge factor $E_X^{(f)}(s)$ and the four diagnostic instruments developed in Papers I–XI [14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24] to $\mathcal{D}_{\mathrm{DH}}(s)$ at (d_*, γ_*), how does the failure of multiplicativity at the composite modes $N \in \{4, 6, 8\}$ quantitatively overcome the $n = 1$ mother wavelet and trigger a simultaneous topological breakdown across Chladni nodal walls, the parity-reduced Wronskian $\widetilde{W}_{\mathrm{DH}}(d,t)$, the Hadamard flux, and the de Branges reproducing kernel [6, 3, 11]?

In this paper (Paper XII), we resolve both questions completely. Our main results are consolidated in the following theorem.

Theorem 1.1 (Universal Selberg-Class Confocal Ellipse Law vs. Davenport–Heilbronn Breakdown)**.** *Let $L(s,f) = \sum_{n=1}^{\infty} a_f(n) n^{-s}$ ($a_f(n) \in \mathbb{R}$, $a_f(1) = 1$) be a self-dual Dirichlet series with completed function $\xi_f(s) = \mathcal{G}_f(s)L(s,f) = \xi_f(1-s)$, where $\mathcal{G}_f(s) = Q_f^s \gamma_f(s)$ has degree $d_f > 0$, Archimedean phase velocity $\vartheta_f'(t) := \frac{\mathrm{d}}{\mathrm{d}t} \arg \mathcal{G}_f(\frac{1}{2} + \mathrm{i}t)$, and saddle horizon $N_f(t) := \lfloor \exp(\vartheta_f'(t)) \rfloor$. Then:*

(i) ***Universal Mother-Wavelet Dilation and Symmetric Branch Damping.*** *The even kernel $\Phi_f(u) = \Phi_f(-u)$ in $\Xi_f(z) = 4\int_{-\infty}^{\infty} \Phi_f(u)\,\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u$ decomposes into dilated shells $\Phi_f(u) = \sum_{m=1}^{\infty} \phi_m^{(f)}(u)$ with $\phi_m^{(f)}(u) \equiv a_f(m) m^{-1/2} \varphi_1^{(f)}(u + \frac{1}{2}\ln m)$, where $\mathcal{G}_f(s) = 4\int_{-\infty}^{\infty} \varphi_1^{(f)}(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w$. Across all active modes $1 \le n \le N_f(t)$ ($\omega_{n,Q_f}(t) := \vartheta_f'(t) - \ln n \ge 0$), strict trigamma convexity enforces*

$$\rho_{\Gamma,f}(d,t)\,|a_f(n)|\,n^{-1/2+d} < \mathrm{e}^{-2d\omega_{n,Q_f}(t)}\,|a_f(n)|\,n^{-1/2-d} \le |a_f(n)|\,n^{-1/2-d} \qquad (d > 0). \tag{1.3}$$

(ii) ***Universal Phase-Independent Confocal Ellipse Identity.*** *For every combined forward and reflected mode $T_n^{(f)}(d,t) := 2a_n n^{-1/2}(\cosh(d\omega_{n,Q}(t))\cos\phi_n(t) + \mathrm{i}\sinh(d\omega_{n,Q}(t))\sin\phi_n(t))$ ($\phi_n(t) = \vartheta_f(t) - t\ln n$), regardless of multiplicativity or an Euler product,*

$$\left|T_n^{(f)}(d,t)\right|^2 - \left|T_n^{(f)}(0,t)\right|^2 \equiv 4\,\frac{|a_n|^2}{n}\,\sinh^2(d\omega_{n,Q}(t)) \ge 0 \qquad (\forall\,\phi_n(t) \in \mathbb{R}), \tag{1.4}$$

tracing a confocal ellipse $\mathcal{E}_n^{(f)}(d,t)$ with d-invariant focal distance $c_n^{(f)} \equiv 2|a_n| n^{-1/2}$, symplectic area $2\pi\frac{|a_n|^2}{n}\sinh(2d|\omega_{n,Q}(t)|)$, and relative increment ratio bounded by $|a_n|^2 n^{-(1+2d)}$.

(iii) ***Multiplicativity Defect $\pm(1+\kappa^2)$ and Prime-Gauge Breakdown in $\mathcal{D}_{\mathrm{DH}}(s)$.*** *Whereas any Euler-product L-function satisfies quadratic prime-power purity $b_N^{\mathcal{E},(f)}(X) = \frac{\Lambda_f(N)}{\ln N}$ on $(X, X^2]$, $\mathcal{D}_{\mathrm{DH}}(s)$ suffers an exact algebraic multiplicativity defect $\delta_{\mathrm{DH}}(m,n) := a_{\mathrm{DH}}(mn) -$*

$a_{\mathrm{DH}}(m)a_{\mathrm{DH}}(n) = \mp(1+\kappa^2) \neq 0$ *at* $N = 4 = 2^2$, $N = 6 = 2\cdot 3$, $N = 9 = 3^2$, *and* $N = 14 = 2\cdot 7$, *leaving unsieved composite residuals* $b_4^{\mathcal{E}}(X) = -(1+\kappa^2)$ *and* $b_6^{\mathcal{E}}(X) = +(1+\kappa^2)$. *At* $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$ *(*$N_5(\gamma_*) = 8$*), all active primes* $p \in \{2,3,5,7\}$ *supply only* 38.82% *of* $|T_1^{(\mathrm{DH})}|$ *(incapable of creating a zero), whereas the composites* $n \in \{4,6,8\}$ *(enhanced by* $1/\kappa^2 = 12.3914\times$*) supply the remaining* 54.03% *destructive interference.*

(iv) ***Four-Fold Diagnostic Detection of*** (d_*, γ_*)**.** *At* (d_*, γ_*)*, (a) critical-line saddle recombination creates a false extremum* $z_c = \tau_{\mathrm{false}} = 85.7472008265$ *where* $\Xi_{\mathrm{DH}}(z_c) < 0$ *and* $\Xi''_{\mathrm{DH}}(z_c) < 0$ *share the same sign, reversing Cauchy–Riemann wall monotonicity* $\frac{\mathrm{d}|R_{\mathrm{DH}}|}{\mathrm{d}s}\big|_{\Gamma_{\mathrm{false}}} = -|\nabla I_{\mathrm{DH}}| < 0$ *until collision with the detached parabola* $R_{\mathrm{DH}} = 0$ *(*$d_{\mathrm{tip}} = 0.2980980120$*); (b)* $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*) < 0$ *on* $[0, d_*)$ *with* $\int_0^{d_*} u\widetilde{W}_{\mathrm{DH}}(u,\gamma_*)\,\mathrm{d}u = -\frac{1}{2}|\Xi_{\mathrm{DH}}(\gamma_*)|^2 < 0$*; (c) the Hadamard flux jumps by* $\Phi_{\mathrm{Had}}(d; d_*) = \pi\,\Theta(d-d_*)$*; and (d) the de Branges inner function crosses* $|\Theta_\alpha(\gamma_* + \mathrm{i}d_*)| = 1$ *with Sylvester inertia jump* $\nu_- : 0 \to 1$.

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned} \kappa &= 0.2840790438404122\ldots, & 1+\kappa^2 &= 1.0807009031492828\ldots,\\ d_* &= 0.3085171824566373\ldots, & \gamma_* &= 85.6993484853775921\ldots,\\ z_c = \tau_{\mathrm{false}} &= 85.7472008264612972\ldots, & d_{\mathrm{tip}} &= 0.2980980119818233\ldots \end{aligned} \tag{1.5}$$

1.1. **Proof overview.** The proof of Theorem 1.1 combines three quantitative ingredients developed in Sections 2–5:

(1) *Bilateral Laplace–Mellin mother-wavelet inversion and single-mode hyperbolic trigonometry (Sections 2 and 3).* Representing the Archimedean prefactor $\mathcal{G}_f(s) = Q_f^s\gamma_f(s)$ as a bilateral Laplace transform in $w = u + \frac{1}{2}\ln m$, we factor out $a_f(m)m^{-1/2}\mathrm{e}^{-2(s-1/2)\frac{1}{2}\ln m} = a_f(m)m^{-s}$ for every degree-d_f self-dual L-function, yielding closed-form mother wavelets for $\zeta(s)$, odd Dirichlet/Davenport–Heilbronn functions, and Ramanujan's modular cusp form $L(s,\Delta)$ [7]. Applying $\cosh^2(d\omega_{n,Q}) - 1 \equiv \sinh^2(d\omega_{n,Q})$ and $\cos^2\phi_n + \sin^2\phi_n \equiv 1$ to each mode $T_n^{(f)}(d,t)$ then establishes (1.4) independently of multiplicativity.

(2) *Exact algebraic multiplicativity defect* $\delta_{\mathrm{DH}}(m,n) = \mp(1+\kappa^2)$ *and prime-gauge convolution analysis (Section 4).* Expanding the product of $\mathcal{D}_{\mathrm{DH}}(s)$ with either the finite Euler gauge $\prod_{p\le X}(1 - a_{\mathrm{DH}}(p)p^{-s})$ or the zero-free exponential prime-power gauge $E_X^{\mathcal{E}}(s) = \exp(-\sum_{p^k\le X}\frac{a_{\mathrm{DH}}(p)^k}{k}p^{-ks})$, we prove that the non-multiplicative jump from $|a_{\mathrm{DH}}(2)| = |a_{\mathrm{DH}}(3)| = \kappa$ to $|a_{\mathrm{DH}}(4)| = |a_{\mathrm{DH}}(6)| = 1$ leaves exact residuals $b_4^{\mathcal{E}}(X) = -(1+\kappa^2)$ and $b_6^{\mathcal{E}}(X) = +(1+\kappa^2)$, amplifying the composite modes $n \in \{4,6\}$ by $1/\kappa^2 = 12.3914\times$ at (d_*, γ_*).

(3) *Cauchy–Riemann saddle bifurcation, Hadamard partial fractions, and Pontryagin–Krein signature jump (Section 5).* Analyzing the critical-line curvature $\partial_d^2 R_{\mathrm{DH}}(0, z_c) = -\Xi''_{\mathrm{DH}}(z_c)$ at the false critical point $z_c = \tau_{\mathrm{false}}$, we show that $\Xi_{\mathrm{DH}}(z_c)\Xi''_{\mathrm{DH}}(z_c) > 0$ reverses the sign of $\frac{\mathrm{d}|R_{\mathrm{DH}}|}{\mathrm{d}s}$ along Γ_{false}, forcing $|R_{\mathrm{DH}}|$ to decrease to zero at (d_*, γ_*), where the mirror Hadamard kernel $K(d, d_*, u)$ undergoes the quantized step jump $\pi\,\Theta(d-d_*)$ and the de Branges Pick–Nevanlinna matrix transitions from $(3,0,0)$ to $(2,0,1)$.

Together, Ingredients (1)–(3) demonstrate that single-mode symplectic convexity holds throughout the Selberg class, whereas global transverse closure across the critical strip relies decisively on the Euler product to suppress composite cross-modes in $[2, N_f(t)]$.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with $d = \sigma - \frac{1}{2} \ge 0$ and $t > 0$, and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. For any self-dual completed L-function $\xi_f(s) = \xi_f(1-s)$ with real coefficients on the critical line, we write $\Xi_f(z) := \xi_f(\frac{1}{2} + \mathrm{i}z) = R_f(d,t) + \mathrm{i}\,I_f(d,t)$, where $R_f(-d,t) = R_f(d,t)$ and $I_f(-d,t) = -I_f(d,t)$. Because $z = t - \mathrm{i}d$ has imaginary part

$-d$, the Cauchy–Riemann equations for $\Xi_f(z)$ read $\partial_d R_f = +\partial_t I_f$ and $\partial_d I_f = -\partial_t R_f$. The transverse Wronskian is $W_f(d,t) := \frac{1}{2}\partial_d|\Xi_f(t-\mathrm{i}d)|^2 = R_f\,\partial_d R_f + I_f\,\partial_d I_f = R_f\,\partial_t I_f - I_f\,\partial_t R_f$, and the parity-reduced Wronskian is $\widetilde{W}_f(d,t) := W_f(d,t)/d$. Implied constants in $\mathcal{O}(\cdot)$ depend only on the Archimedean parameters (Q_f, λ_j, μ_j) unless stated otherwise. All high-precision numerical values are computed at 60-decimal working precision (`mpmath.mp.dps = 60`) and certified via 35-decimal directed-rounding interval arithmetic (`mpmath.iv.dps = 35`) in `scripts/verify_paper12_selberg_davenport_heilbronn.py`.

2. Universal Mother-Wavelet Dilation and Symmetric Branch Damping in the Selberg Class

Let $L(s,f) = \sum_{n=1}^{\infty} a_f(n) n^{-s}$ be a Dirichlet series with real coefficients $a_f(n) \in \mathbb{R}$ ($a_f(1) = 1$, $a_f(n) = \mathcal{O}(n^{\varepsilon})$) whose completed function

$$\xi_f(s) := \mathcal{G}_f(s)\,L(s,f) = \xi_f(1-s), \qquad \mathcal{G}_f(s) := Q_f^s\,P_f(s)\prod_{j=1}^{r}\Gamma(\lambda_j s + \mu_j), \tag{2.1}$$

satisfies a self-dual functional equation of degree $d_f := 2\sum_{j=1}^{r}\lambda_j > 0$, conductor $Q_f > 0$, spectral shifts $\mu_j \in \mathbb{R}$ with $\frac{1}{2}\lambda_j + \mu_j > 0$, and polynomial factor $P_f(s) = P_f(1-s)$ [27]. We consider four canonical representatives across degrees $d_f \in \{1,2\}$:

(a) *Riemann's zeta function $\zeta(s)$ ($d_\zeta = 1$):* $a_\zeta(n) = 1$, $Q_\zeta = \pi^{-1/2}$, $P_\zeta(s) = \frac{1}{2}s(s-1)$, and $\mathcal{G}_\zeta(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$.

(b) *Odd primitive self-dual combinations modulo q ($d_q = 1$, $q = 5$):* Both the rotated odd Dirichlet L-function and the Davenport–Heilbronn function $\mathcal{D}_{\mathrm{DH}}(s)$ in (1.1) share the identical Archimedean factor $\mathcal{G}_5(s) = (\frac{\pi}{5})^{-(s+1)/2}\Gamma(\frac{s+1}{2})$ with effective conductor $Q_5 = (5/\pi)^{1/2}$.

(c) *Ramanujan's modular cusp form $L(s,\Delta)$ of weight $k = 12$ for $\mathrm{SL}_2(\mathbb{Z})$ ($d_\Delta = 2$):* Here $\Delta(\tau) = q\prod_{m=1}^{\infty}(1-q^m)^{24} = \sum_{n=1}^{\infty}\tau(n)q^n$ ($q = \mathrm{e}^{2\pi\mathrm{i}\tau}$), the analytically normalized real Hecke coefficients are $a_\Delta(n) := \tau(n)n^{-11/2} \in \mathbb{R}$ (satisfying Deligne's Ramanujan–Petersson bound $|a_\Delta(p)| \le 2$ [7]), and the completed L-function is $\xi_\Delta(s) = \mathcal{G}_\Delta(s)L(s,\Delta) = \xi_\Delta(1-s)$ with $\mathcal{G}_\Delta(s) = (2\pi)^{-(s+11/2)}\Gamma(s+\frac{11}{2})$ and $Q_\Delta = (2\pi)^{-1}$.

Before exchanging summation and bilateral Laplace–Mellin integration, we establish the holomorphic decay of the class mother wavelet. For any vertical strip $\sigma \in [\sigma_1,\sigma_2]$ free of poles of $\mathcal{G}_f(s)$, Stirling's asymptotic formula gives $|\mathcal{G}_f(\sigma+\mathrm{i}t)| \asymp |t|^{\sum_{j=1}^{r}(\lambda_j\sigma+\mu_j-1/2)+\deg P_f}\exp(-\frac{\pi d_f}{4}|t|)$ as $|t|\to\infty$. Consequently, the inverse bilateral Laplace–Mellin integral

$$\varphi_1^{(f)}(w) := \frac{1}{2\pi\mathrm{i}}\int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty}\mathcal{G}_f(s)\,\mathrm{e}^{-2(s-1/2)w}\,\mathrm{d}s \iff \mathcal{G}_f(s) = 4\int_{-\infty}^{\infty}\varphi_1^{(f)}(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w \tag{2.2}$$

converges absolutely and defines a holomorphic function of $w = \xi+\mathrm{i}\eta$ throughout the strip $|\mathrm{Im}(w)| < \frac{\pi d_f}{8}$, with double-exponential decay as $\xi\to+\infty$ and exponential decay as $\xi\to-\infty$. This absolute convergence justifies term-by-term Mellin inversion for $\sigma > 1$.

Theorem 2.1 (Universal Mother-Wavelet Dilation and Symmetric Branch Damping)**.** *Let $\xi_f(s) = \mathcal{G}_f(s)L(s,f) = \xi_f(1-s)$ be a self-dual completed L-function of the form (2.1). Then:*

(i) ***Exact Logarithmic Shell Dilation.*** *The completed function $\Xi_f(z) = \xi_f(\frac{1}{2}+\mathrm{i}z)$ admits the Fourier–Laplace integral representation $\Xi_f(z) = 4\int_{-\infty}^{\infty}\Phi_f(u)\,\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u = 8\int_0^{\infty}\Phi_f(u)\cos(2zu)\,\mathrm{d}u$, where the even kernel $\Phi_f(u) = \Phi_f(-u)$ decomposes into dilated, amplitude-weighted copies of the single class mother wavelet $\varphi_1^{(f)}(w)$:*

$$\Phi_f(u) = \sum_{m=1}^{\infty}\phi_m^{(f)}(u), \qquad \phi_m^{(f)}(u) \equiv \frac{a_f(m)}{\sqrt{m}}\,\varphi_1^{(f)}\Big(u+\frac{1}{2}\ln m\Big) \qquad (\forall\, m\ge 1,\ |\mathrm{Im}(u)| < \tfrac{\pi d_f}{8}). \tag{2.3}$$

In particular, the closed-form mother wavelets and m-th dilated shells for $\zeta(s)$, the odd modulo-q class ($L(s,\chi_5)$ and $\mathcal{D}_{\mathrm{DH}}(s)$ at $q=5$), and Ramanujan's $L(s,\Delta)$ are given explicitly by

$$\begin{aligned}
\varphi_1^{(\zeta)}(w) &= (2\pi^2 \mathrm{e}^{9w} - 3\pi \mathrm{e}^{5w})\mathrm{e}^{-\pi \mathrm{e}^{4w}}, & \phi_m^{(\zeta)}(u) &= (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u})\mathrm{e}^{-\pi m^2 \mathrm{e}^{4u}},\\
\varphi_1^{(q)}(w) &= \mathrm{e}^{3w}\mathrm{e}^{-\frac{\pi}{q}\mathrm{e}^{4w}}, & \phi_m^{(q)}(u) &= a_f(m)\, m\, \mathrm{e}^{3u}\mathrm{e}^{-\frac{\pi}{q} m^2 \mathrm{e}^{4u}},\\
\varphi_1^{(\Delta)}(w) &= \frac{1}{2}\mathrm{e}^{12w}\mathrm{e}^{-2\pi \mathrm{e}^{2w}}, & \phi_m^{(\Delta)}(u) &= \frac{1}{2}\tau(m)\, \mathrm{e}^{12u}\mathrm{e}^{-2\pi m \mathrm{e}^{2u}}.
\end{aligned} \tag{2.4}$$

(ii) ***Universal Symmetric Branch-Damping Inequality.*** *Let $\vartheta_f(t) := \arg \mathcal{G}_f(\frac{1}{2}+\mathrm{i}t)$, let $\vartheta_f'(t) = \ln Q_f + \sum_{j=1}^r \lambda_j \operatorname{Re}\psi(\frac{1}{2}\lambda_j + \mu_j + \mathrm{i}\lambda_j t) + \operatorname{Re}\frac{P_f'}{P_f}(\frac{1}{2}+\mathrm{i}t) \sim \frac{d_f}{2}\ln t$ be the Archimedean phase velocity, and let $\omega_{n,Q_f}(t) := \vartheta_f'(t) - \ln n = \ln(N_\star^{(f)}(t)/n)$ with $N_\star^{(f)}(t) := \exp(\vartheta_f'(t)) \asymp Q_f t^{d_f/2}$. Then for all t above the critical inflection height where $\partial_u^2 \ln|\mathcal{G}_f(\frac{1}{2}+u+\mathrm{i}t)| > 0$ on $u \in (0,d]$, the Archimedean branch ratio $\rho_{\Gamma,f}(d,t) := |\mathcal{G}_f(\frac{1}{2}-d-\mathrm{i}t)/\mathcal{G}_f(\frac{1}{2}+d+\mathrm{i}t)|$ satisfies $\rho_{\Gamma,f}(d,t) < \mathrm{e}^{-2d\vartheta_f'(t)} < 1$, and every active mode $1 \le n \le N_f(t) := \lfloor N_\star^{(f)}(t)\rfloor$ ($\omega_{n,Q_f}(t) \ge 0$) obeys*

$$\rho_{\Gamma,f}(d,t)\,|a_f(n)|\,n^{-1/2+d} < \mathrm{e}^{-2d\omega_{n,Q_f}(t)}\,|a_f(n)|\,n^{-1/2-d} \le |a_f(n)|\,n^{-1/2-d} \qquad (d>0). \tag{2.5}$$

Proof. (i) For $\operatorname{Re}(s) > 1$, substituting $w = u + \frac{1}{2}\ln m$ ($\mathrm{d}w = \mathrm{d}u$) into the identity $\mathcal{G}_f(s) = 4\int_{-\infty}^{\infty} \varphi_1^{(f)}(w)\,\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w$ multiplies the integral by $\mathrm{e}^{-(s-1/2)\ln m} = m^{1/2-s}$. Multiplying by $a_f(m)m^{-1/2}$ and summing over $m \ge 1$ via Fubini's theorem (justified by the double-exponential decay $\exp(-Cm^{2/d_f}\mathrm{e}^{4u/d_f})$ for $u \gg 1$) yields

$$4\int_{-\infty}^{\infty}\left(\sum_{m=1}^{\infty}\frac{a_f(m)}{\sqrt{m}}\,\varphi_1^{(f)}\!\left(u+\frac{1}{2}\ln m\right)\right)\mathrm{e}^{2(s-1/2)u}\,\mathrm{d}u = \mathcal{G}_f(s)\sum_{m=1}^{\infty} a_f(m)\,m^{-s} = \xi_f(s).$$

Since $\xi_f(s) = \xi_f(1-s)$ is invariant under $s-\frac{1}{2} \mapsto -(s-\frac{1}{2})$, uniqueness of the bilateral Laplace transform implies $\Phi_f(-u) = \Phi_f(u)$ identically. For the odd modulo-q class, substituting $x = \frac{\pi}{q}\mathrm{e}^{4w}$ ($\mathrm{d}w = \frac{\mathrm{d}x}{4x}$) gives $4\int_{-\infty}^{\infty}\mathrm{e}^{3w}\mathrm{e}^{-\frac{\pi}{q}\mathrm{e}^{4w}}\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = (\frac{\pi}{q})^{-(s+1)/2}\int_0^\infty x^{(s+1)/2-1}\mathrm{e}^{-x}\,\mathrm{d}x = \mathcal{G}_{\mathrm{odd},q}(s)$, and $\frac{a_f(m)}{\sqrt{m}}\mathrm{e}^{3(u+\frac{1}{2}\ln m)} = a_f(m)m\,\mathrm{e}^{3u}$. For Ramanujan's Δ, substituting $x = 2\pi\mathrm{e}^{2w}$ ($\mathrm{d}w = \frac{\mathrm{d}x}{2x}$) gives $4\int_{-\infty}^{\infty}\frac{1}{2}\mathrm{e}^{12w}\mathrm{e}^{-2\pi\mathrm{e}^{2w}}\mathrm{e}^{2(s-1/2)w}\,\mathrm{d}w = (2\pi)^{-(s+11/2)}\Gamma(s+\frac{11}{2}) = \mathcal{G}_\Delta(s)$, while $\frac{a_\Delta(m)}{\sqrt{m}}\mathrm{e}^{12(u+\frac{1}{2}\ln m)} = \tau(m)m^{-6}m^6\mathrm{e}^{12u} = \tau(m)\mathrm{e}^{12u}$, so $\Phi_\Delta(u) = \frac{1}{2}\mathrm{e}^{12u}\Delta(\mathrm{i}\mathrm{e}^{2u})$.

(ii) By Cauchy–Riemann on $\ln\mathcal{G}_f(\frac{1}{2}+u+\mathrm{i}t)$, $\partial_u \ln|\mathcal{G}_f(\frac{1}{2}+u+\mathrm{i}t)|_{u=0} = \partial_t \arg\mathcal{G}_f(\frac{1}{2}+\mathrm{i}t) = \vartheta_f'(t)$ (note that any real self-dual polynomial factor $P_f(1-s) = P_f(s)$ satisfies $|P_f(\frac{1}{2}-u-\mathrm{i}t)| \equiv |P_f(\frac{1}{2}+u+\mathrm{i}t)|$, so P_f cancels identically from $\rho_{\Gamma,f}(d,t)$ and $\operatorname{Re}\frac{P_f'}{P_f}(\frac{1}{2}+\mathrm{i}t) \equiv 0$). Define the symmetrized logarithmic slope $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t) := \partial_u \ln|\mathcal{G}_f(\frac{1}{2}+u+\mathrm{i}t)| + \partial_u\ln|\mathcal{G}_f(\frac{1}{2}-u+\mathrm{i}t)|$. At $u=0$, $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(0,t) = 2\vartheta_f'(t)$, while $\partial_u\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(0,t) = 0$ by evenness. Because $\partial_u^2\ln|\mathcal{G}_f(\frac{1}{2}\pm u+\mathrm{i}t)| = \sum_{j=1}^r\lambda_j^2\operatorname{Re}\psi_1(\frac{1}{2}\lambda_j+\mu_j \pm \lambda_j u + \mathrm{i}\lambda_j t) + \operatorname{Re}(\frac{P_f'}{P_f})' > 0$ for large t (since $\operatorname{Re}\psi_1(\sigma+\mathrm{i}\tau) = \tau^{-2}\sigma + \mathcal{O}(\tau^{-3}) > 0$), $\mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t)$ is strictly convex on $u \in (0,d]$, so $\int_0^d \mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t)\,\mathrm{d}u > 2d\vartheta_f'(t)$. Exponentiation gives $\rho_{\Gamma,f}(d,t) = \exp(-\int_0^d \mathcal{A}_{\Gamma,f}^{\mathrm{sym}}(u,t)\,\mathrm{d}u) < \mathrm{e}^{-2d\vartheta_f'(t)}$. Multiplying by $|a_f(n)|n^{-1/2+d} = |a_f(n)|n^{-1/2-d}\mathrm{e}^{2d\ln n}$ yields $\rho_{\Gamma,f}(d,t)|a_f(n)|n^{-1/2+d} < \mathrm{e}^{-2d(\vartheta_f'(t)-\ln n)}|a_f(n)|n^{-1/2-d} = \mathrm{e}^{-2d\omega_{n,Q_f}(t)}|a_f(n)|n^{-1/2-d} \le |a_f(n)|n^{-1/2-d}$ whenever $\omega_{n,Q_f}(t) \ge 0$. □

Table 1 records the high-precision verification of Theorem 2.1 across $\zeta(s)$, $L(s,\chi_5)$, $\mathcal{D}_{\mathrm{DH}}(s)$, and $L(s,\Delta)$.

TABLE 1. Verification of the bilateral Laplace–Mellin mother-wavelet identity (2.4) and the symmetric branch-damping inequality (2.5) at $d = d_* = 0.3085171825$ across degrees $d_f \in \{1, 2\}$.

L-Function Class	Deg. d_f	Height t	$N_\star^{(f)}(t)$	$N_f(t)$	$\rho_{\Gamma,f}(d_*, t)$	$\mathrm{e}^{-2d_*\vartheta_f'(t)}$	Mellin Rel. Err
Riemann $\zeta(s)$	1	85.699348	3.6931555	3	0.4465765317	0.4465768293	1.28×10^{-61}
Dirichlet $L(s, \chi_5)$	1	85.699348	8.2581468	8	0.2718013692	0.2718015503	1.33×10^{-61}
Davenport–Heilbronn $\mathcal{D}_{\mathrm{DH}}(s)$	1	85.699348	8.2581468	8	0.2718013692	0.2718015503	1.33×10^{-61}
Ramanujan cusp form $L(s, \Delta)$	2	25.000000	4.0737894	4	0.4203432003	0.4203489013	1.84×10^{-61}

3. The Universal Phase-Independent Confocal Ellipse Identity

In the Riemann–Siegel approximate functional equation for $\Xi_f(t - \mathrm{i}d)$ [28, 31], factoring out the Archimedean envelope $(N_\star^{(f)}(t))^{-d}$ pairs the n-th forward Dirichlet term $a_n n^{-1/2-d-\mathrm{i}t}\mathrm{e}^{\mathrm{i}\vartheta_f(t)}(N_\star^{(f)}(t))^d = a_n n^{-1/2}\mathrm{e}^{d\omega_{n,Q}(t)}\mathrm{e}^{\mathrm{i}\phi_n(t)}$ with its reflected counterpart $a_n n^{-1/2+d+\mathrm{i}t}\mathrm{e}^{-\mathrm{i}\vartheta_f(t)}(N_\star^{(f)}(t))^{-d} = a_n n^{-1/2}\mathrm{e}^{-d\omega_{n,Q}(t)}\mathrm{e}^{-\mathrm{i}\phi_n(t)}$, where $\omega_{n,Q}(t) := \vartheta_f'(t) - \ln n$ and $\phi_n(t) := \vartheta_f(t) - t\ln n$. We now prove that the transverse modulus growth of every combined mode $T_n^{(f)}(d, t)$ is completely independent of its oscillating phase $\phi_n(t)$, regardless of whether $L(s, f)$ has an Euler product.

Theorem 3.1 (Universal Phase-Independent Confocal Ellipse Identity)**.** *Let $L(s, f) = \sum_{n=1}^\infty a_n n^{-s}$ be any Dirichlet series with real coefficients $a_n \in \mathbb{R}$ ($a_1 = 1$), saddle velocity $\omega_{n,Q}(t) = \vartheta_f'(t) - \ln n$, and phase $\phi_n(t) = \vartheta_f(t) - t\ln n \in \mathbb{R}$. For $d \geq 0$, define the combined n-th Riemann–Siegel mode*

$$\begin{aligned} T_n^{(f)}(d, t) &:= \frac{a_n}{\sqrt{n}}\Big(\mathrm{e}^{d\omega_{n,Q}(t)}\mathrm{e}^{\mathrm{i}\phi_n(t)} + \mathrm{e}^{-d\omega_{n,Q}(t)}\mathrm{e}^{-\mathrm{i}\phi_n(t)}\Big) \\ &= 2\,\frac{a_n}{\sqrt{n}}\Big(\cosh\big(d\omega_{n,Q}(t)\big)\cos\phi_n(t) + \mathrm{i}\sinh\big(d\omega_{n,Q}(t)\big)\sin\phi_n(t)\Big). \end{aligned} \tag{3.1}$$

Then for every integer $n \geq 1$, every $d \geq 0$, and every $t > 0$, the following four exact identities hold:

(i) ***Phase-Independent Transverse Modulus Increment.*** *The difference between $|T_n^{(f)}(d, t)|^2$ off the critical line and $|T_n^{(f)}(0, t)|^2 = 4\frac{|a_n|^2}{n}\cos^2\phi_n(t)$ on the critical line is independent of $\phi_n(t)$:*

$$\boxed{\Big|T_n^{(f)}(d, t)\Big|^2 - \Big|T_n^{(f)}(0, t)\Big|^2 \equiv 4\,\frac{|a_n|^2}{n}\,\sinh^2\big(d\omega_{n,Q}(t)\big) \geq 0 \qquad (\forall\,\phi_n(t) \in \mathbb{R}).} \tag{3.2}$$

(ii) ***d-Invariant Confocal Foci and Symplectic Area.*** *As $\phi_n \in [0, 2\pi)$ varies at fixed (d, t) with $d > 0$ and $\omega_{n,Q}(t) \neq 0$, $T_n^{(f)}(d, t)$ traces a confocal ellipse $\mathcal{E}_n^{(f)}(d, t)$ in $\mathbb{C}$ with semi-major axis $A_n(d, t) = 2|a_n|n^{-1/2}\cosh(d\omega_{n,Q}(t))$ and semi-minor axis $B_n(d, t) = 2|a_n|n^{-1/2}\sinh(d|\omega_{n,Q}(t)|)$, whose linear focal distance $c_n^{(f)}$ is invariant under d and t and whose enclosed symplectic area reproduces the odd Wigner–sinh weight:*

$$c_n^{(f)} := \sqrt{A_n(d, t)^2 - B_n(d, t)^2} \equiv 2\,\frac{|a_n|}{\sqrt{n}}, \qquad \mathrm{Area}(\mathcal{E}_n^{(f)}(d, t)) = \pi A_n B_n \equiv 2\pi\,\frac{|a_n|^2}{n}\,\sinh\big(2d|\omega_{n,Q}(t)|\big). \tag{3.3}$$

(iii) ***Universal $|a_n|^2 n^{-(1+2d)}$ Diagonal Ratio Law.*** *Relative to the $n = 1$ mother-wavelet mode ($a_1 = 1$, $\omega_{1,Q}(t) = \ln N_\star^{(f)}(t)$), every active mode $1 \leq n \leq N_\star^{(f)}(t)$ satisfies*

$$\frac{|T_n^{(f)}(d, t)|^2 - |T_n^{(f)}(0, t)|^2}{|T_1^{(f)}(d, t)|^2 - |T_1^{(f)}(0, t)|^2} \equiv |a_n|^2\,n^{-(1+2d)}\left(\frac{1 - (n/N_\star^{(f)}(t))^{2d}}{1 - (N_\star^{(f)}(t))^{-2d}}\right)^2 \leq |a_n|^2\,n^{-(1+2d)}. \tag{3.4}$$

TABLE 2. Mode-by-mode verification of the universal confocal ellipse identity (3.2) across all $N_5(\gamma_*) = 8$ active modes of $\mathcal{D}_{\mathrm{DH}}(s)$ at $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$ ($N_\star^{(\mathrm{DH})}(\gamma_*) = 8.25814675$).

Mode n	Type	$a_{\mathrm{DH}}(n)$	$\mathrm{Re}\, T_n^{(\mathrm{DH})}(d_*, \gamma_*)$	$\mathrm{Im}\, T_n^{(\mathrm{DH})}(d_*, \gamma_*)$	$\lvert T_n(d_*)\rvert^2 - \lvert T_n(0)\rvert^2$	Focus $c_n^{(\mathrm{DH})}$	60-dps Err
1	Mother wavelet	$+1.000000000$	$+2.368423174$	$+0.3346136251$	$+1.9509564293$	2.000000000	7.78×10^{-61}
2	Prime	$+0.284079044$	-0.380355713	-0.0917020609	$+0.0329143647$	0.401748437	4.86×10^{-63}
3	Prime	-0.284079044	-0.324537715	-0.0346678847	$+0.0108473550$	0.328026225	4.86×10^{-63}
4	Composite 2^2	-1.000000000	-0.700916596	-0.1645628715	$+0.0508569892$	1.000000000	4.86×10^{-63}
5	Prime $(5 \mid q)$	0.000000000	0.000000000	0.0000000000	0.0000000000	0.000000000	0.00×10^{0}
6	Composite $2 \cdot 3$	$+1.000000000$	-0.664243313	-0.0473105465	$+0.0064961139$	0.816496581	8.75×10^{-62}
7	Prime	$+0.284079044$	-0.214992124	$+0.0001852011$	$+0.0001200242$	0.214743572	1.59×10^{-63}
8	Composite 2^3	-0.284079044	$+0.090060189$	$+0.0017593352$	$+0.0000038739$	0.200874218	1.74×10^{-63}

(iv) ***Oscillation-Free Diagonal Wronskian and $\mathcal{O}(N_f(t)^{-3})$ Boundary Self-Damping.*** *The diagonal Wronskian sum* $\mathcal{D}_{\mathrm{diag}}^{(f)}(d,t) := \sum_{n=1}^{N_f(t)} \frac{1}{2d}\partial_d |T_n^{(f)}(d,t)|^2 = \sum_{n=1}^{N_f(t)} \frac{4|a_n|^2}{n} \omega_{n,Q}(t)^2 \mathrm{sinc}_{\mathrm{hyp}}(2d\omega_{n,Q}(t)) > 0$ *(*$\mathrm{sinc}_{\mathrm{hyp}} x := \frac{\sinh x}{x}$*) is strictly positive and phase-free, while the slowest adjacent boundary cross-pair* $(N_f - 1, N_f)$ *self-damps at rate* $\frac{(\omega_{N_f-1,Q} + \omega_{N_f,Q})^2}{\sqrt{(N_f-1)N_f}} = \mathcal{O}(N_f(t)^{-3}) = \mathcal{O}(t^{-3d_f/2}) \to 0$.

Proof. (i) Since $a_n \in \mathbb{R}$ and $\phi_n(t) \in \mathbb{R}$, separating (3.1) into real and imaginary parts gives $\mathrm{Re}\, T_n^{(f)}(d,t) = 2a_n n^{-1/2} \cosh(d\omega_{n,Q}) \cos\phi_n$ and $\mathrm{Im}\, T_n^{(f)}(d,t) = 2a_n n^{-1/2} \sinh(d\omega_{n,Q}) \sin\phi_n$. Substituting $\cosh^2(d\omega_{n,Q}) \equiv 1 + \sinh^2(d\omega_{n,Q})$ into $|T_n^{(f)}(d,t)|^2 = (\mathrm{Re}\, T_n^{(f)})^2 + (\mathrm{Im}\, T_n^{(f)})^2$ yields

$$\begin{aligned} \left|T_n^{(f)}(d,t)\right|^2 &= 4\,\frac{|a_n|^2}{n}\Big[\big(1 + \sinh^2(d\omega_{n,Q})\big)\cos^2\phi_n + \sinh^2(d\omega_{n,Q})\sin^2\phi_n\Big] \\ &= 4\,\frac{|a_n|^2}{n}\cos^2\phi_n + 4\,\frac{|a_n|^2}{n}\sinh^2(d\omega_{n,Q}(t))\big(\cos^2\phi_n + \sin^2\phi_n\big). \end{aligned}$$

Since $|T_n^{(f)}(0,t)|^2 = 4\frac{|a_n|^2}{n}\cos^2\phi_n$ and $\cos^2\phi_n + \sin^2\phi_n \equiv 1$, subtracting $|T_n^{(f)}(0,t)|^2$ proves (3.2).

(ii) Using $\cosh^2(d\omega_{n,Q}) - \sinh^2(d\omega_{n,Q}) \equiv 1$ gives $A_n^2 - B_n^2 = 4\frac{|a_n|^2}{n}$, while $\pi A_n B_n = 4\pi\frac{|a_n|^2}{n}\cosh(d\omega_{n,Q})\sinh(d|\omega_{n,Q}|) = 2\pi\frac{|a_n|^2}{n}\sinh(2d|\omega_{n,Q}|)$.

(iii) Since $\mathrm{e}^{\omega_{n,Q}(t)} = N_\star^{(f)}(t)/n$, we have $\sinh(d\omega_{n,Q}(t)) = \frac{1}{2}(N_\star^{(f)}(t))^d n^{-d}(1 - (n/N_\star^{(f)}(t))^{2d})$. Squaring, multiplying by $4|a_n|^2/n$, and dividing by the $n = 1$ expression yields (3.4).

(iv) Differentiating (3.2) with respect to d gives $\frac{1}{2d}\partial_d|T_n^{(f)}(d,t)|^2 = \frac{2|a_n|^2}{nd}\sinh(2d\omega_{n,Q})\omega_{n,Q} = \frac{4|a_n|^2}{n}\omega_{n,Q}^2\mathrm{sinc}_{\mathrm{hyp}}(2d\omega_{n,Q}) > 0$. At $N_f = \lfloor N_\star^{(f)}(t)\rfloor$, $0 \le \omega_{N_f,Q}(t) < \ln(1 + 1/N_f) < 1/N_f$ and $\omega_{N_f-1,Q}(t) < \frac{3}{2(N_f-1)}$, so $\frac{(\omega_{N_f-1,Q}+\omega_{N_f,Q})^2}{\sqrt{(N_f-1)N_f}} = \mathcal{O}(N_f^{-3})$. □

Table 2 displays the exact mode-by-mode verification of Theorem 3.1 across all $N_5(\gamma_*) = 8$ active modes of $\mathcal{D}_{\mathrm{DH}}(s)$ at its first off-line zero (d_*, γ_*). Every single mode of $\mathcal{D}_{\mathrm{DH}}(s)$ obeys the confocal ellipse identity (3.2) to 60 decimal places (with $0 \in$ [LHS − RHS] certified in 35-digit interval arithmetic).

4. Multiplicativity Defect $\pm(1+\kappa^2)$ and Breakdown of Prime-Gauge Purity in $\mathcal{D}_{\mathrm{DH}}(s)$

Why, then, does $\mathcal{D}_{\mathrm{DH}}(s)$ admit an off-line zero at (d_*, γ_*) despite obeying Theorem 2.1 and Theorem 3.1? For any automorphic L-function $L(s,f) = \prod_p \prod_{j=1}^{d_f}(1 - \alpha_{p,j}p^{-s})^{-1}$ in the Selberg class, the Euler product ensures that $\ln L(s,f) = \sum_{n=2}^{\infty} \frac{\Lambda_f(n)}{\ln n} n^{-s}$ is supported *exclusively on prime*

powers $n = p^k$ ($\Lambda_f(N) \equiv 0$ whenever N is composite and not a prime power). Consequently, multiplying $L(s,f)$ by the zero-free exponential prime-gauge factor [19]

$$E_X^{(f)}(s) := \exp\left(- \sum_{2 \le p^k \le \lfloor X \rfloor} \frac{\Lambda_f(p^k)}{\ln(p^k)}\, p^{-ks} \right), \qquad L(s,f)\, E_X^{(f)}(s) = \sum_{N=1}^{\infty} b_N^{\mathcal{E},(f)}(X)\, N^{-s} \tag{4.1}$$

not only annihilates all modes $2 \le N \le \lfloor X \rfloor$ identically ($b_N^{\mathcal{E},(f)}(X) \equiv 0$), but also enforces *quadratic prime-power purity* $b_N^{\mathcal{E},(f)}(X) \equiv \frac{\Lambda_f(N)}{\ln N}$ across the entire quadratic window $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$, annihilating every non-prime-power composite ($N = 6, 10, 12, 14, 15, \dots$) in $(\lfloor X \rfloor, (\lfloor X \rfloor + 1)^2)$. We now prove that this mechanism fails at the very first composite integers $N = 4 = 2^2$ and $N = 6 = 2{\cdot}3$ in $\mathcal{D}_{\mathrm{DH}}(s)$.

Theorem 4.1 (Multiplicativity Defect $\pm(1+\kappa^2)$ and Breakdown of Prime-Gauge Purity)**.** *For the Davenport–Heilbronn function* $\mathcal{D}_{\mathrm{DH}}(s) = \sum_{n=1}^{\infty} a_{\mathrm{DH}}(n) n^{-s}$ *with* $\kappa = \frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1}$*, define the multiplicativity defect* $\delta_{\mathrm{DH}}(m,n) := a_{\mathrm{DH}}(mn) - a_{\mathrm{DH}}(m) a_{\mathrm{DH}}(n)$ *for* $m, n \ge 2$*. Then:*

(i) ***Exact Algebraic Defect Law*** $\pm(1+\kappa^2)$**.** *For every pair* (m,n) *with* $m, n \equiv 2$ *or* $3 \pmod 5$*, the multiplicativity defect has exact magnitude* $1 + \kappa^2 = 1.080700903149282891 27\ldots \neq 0$*:*

$$\begin{aligned} &\delta_{\mathrm{DH}}(2,2) = a_{\mathrm{DH}}(4) - a_{\mathrm{DH}}(2)^2 = -(1+\kappa^2), \quad \delta_{\mathrm{DH}}(2,3) = a_{\mathrm{DH}}(6) - a_{\mathrm{DH}}(2) a_{\mathrm{DH}}(3) = +(1+\kappa^2), \\ &\delta_{\mathrm{DH}}(3,3) = a_{\mathrm{DH}}(9) - a_{\mathrm{DH}}(3)^2 = -(1+\kappa^2), \quad \delta_{\mathrm{DH}}(2,7) = a_{\mathrm{DH}}(14) - a_{\mathrm{DH}}(2) a_{\mathrm{DH}}(7) = -(1+\kappa^2). \end{aligned} \tag{4.2}$$

Moreover, $1 + \kappa^2 = 0 \iff \kappa = \pm \mathrm{i}$*, at which values* $\mathcal{D}_{\mathrm{DH}}(s)$ *reduces to the multiplicative Dirichlet L-functions* $L(s, \overline{\chi_5})$ *and* $L(s, \chi_5)$*.*

(ii) ***Unsieved Composite Residuals at*** $N = 4$ ***and*** $N = 6$**.** *Under any prime-generated gauge factor of cutoff* $X \ge 5$*—whether the finite Euler gauge* $E_X^{\mathrm{Euler}}(s) := \prod_{p \le X} (1 - a_{\mathrm{DH}}(p) p^{-s})$ *or the exponential prime-power gauge* $E_X^{\mathcal{E}}(s) := \exp(-\sum_{p^k \le X} \frac{a_{\mathrm{DH}}(p)^k}{k} p^{-ks})$*—the prime modes* $p \in \{2,3,5\}$ *are annihilated (*$b_2^{\mathcal{E}} = b_3^{\mathcal{E}} = b_5^{\mathcal{E}} = 0$*), but the composite modes* $N = 4 = 2^2 \le X$ *and* $N = 6 = 2 \cdot 3$ *retain non-zero residuals exceeding* 1 *in magnitude:*

$$b_4^{\mathcal{E}}(X) = b_4^{\mathrm{Euler}}(X) = \delta_{\mathrm{DH}}(2,2) = -(1+\kappa^2), \qquad b_6^{\mathcal{E}}(X) = b_6^{\mathrm{Euler}}(X) = \delta_{\mathrm{DH}}(2,3) = +(1+\kappa^2). \tag{4.3}$$

(iii) ***Non-Vanishing Formal von Mangoldt Coefficients on Composites.*** *In the formal Dirichlet logarithm* $-\frac{\mathcal{D}'_{\mathrm{DH}}(s)}{\mathcal{D}_{\mathrm{DH}}(s)} = \sum_{n=2}^{\infty} \Lambda_{\mathrm{DH}}(n) n^{-s}$*, the formal coefficients fail to vanish on non-prime-power composites in* $(X, X^2]$*: at* $N = 6 = 2 \cdot 3$ *and* $N = 14 = 2 \cdot 7$*, one has* $\frac{\Lambda_{\mathrm{DH}}(6)}{\ln 6} = +(1+\kappa^2) \neq 0$ *and* $\frac{\Lambda_{\mathrm{DH}}(14)}{\ln 14} = -(1+\kappa^2) \neq 0$*.*

(iv) ***Quantitative Composite Dominance at the Off-Line Zero*** (d_*, γ_*)**.** *In the 8-mode Riemann–Siegel main sum at* $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$*, the* $n = 1$ *mother wavelet has modulus* $|T_1^{(\mathrm{DH})}(d_*, \gamma_*)| = 2.39194369$*, whereas all four active primes* $p \in \{2,3,5,7\}$ *combined satisfy*

$$\Big| \sum_{p \in \{2,3,5,7\}} T_p^{(\mathrm{DH})}(d_*, \gamma_*) \Big| = 0.92849988 = 0.388178\, |T_1^{(\mathrm{DH})}(d_*, \gamma_*)| < |T_1^{(\mathrm{DH})}(d_*, \gamma_*)|, \tag{4.4}$$

and therefore cannot cancel $T_1^{(\mathrm{DH})}$*. The remaining* 54.03% *destructive interference is supplied by the non-multiplicative composite modes* $n \in \{4,6,8\}$ *(*$|\sum_{n \in \{4,6,8\}} T_n^{(\mathrm{DH})}(d_*, \gamma_*)| = 1.29229533 = 0.540270\, |T_1^{(\mathrm{DH})}|$*), whose amplitudes at* $n = 4, 6$ *are amplified by* $1/\kappa^2 = 12.391449\times$ *relative to the multiplicative values* $|a_{\mathrm{DH}}(2)|^2 = |a_{\mathrm{DH}}(2) a_{\mathrm{DH}}(3)| = \kappa^2 = 0.080701$*.*

Proof. (i) Since $a_{\mathrm{DH}}(n)$ is 5-periodic with $a_{\mathrm{DH}}(2) = a_{\mathrm{DH}}(7) = \kappa$, $a_{\mathrm{DH}}(3) = -\kappa$, $a_{\mathrm{DH}}(4) = a_{\mathrm{DH}}(9) = a_{\mathrm{DH}}(14) = -1$, and $a_{\mathrm{DH}}(6) = +1$, direct substitution gives $\delta_{\mathrm{DH}}(2,2) = -1 - \kappa^2$, $\delta_{\mathrm{DH}}(2,3) = 1 - (-\kappa^2) = 1 + \kappa^2$, and $\delta_{\mathrm{DH}}(3,3) = \delta_{\mathrm{DH}}(2,7) = -1 - \kappa^2$.

TABLE 3. Comparison of sieved Dirichlet coefficients at cutoff $X = 5$ between the Euler-product zeta function $\zeta(s)E_5(s)$ and the Davenport–Heilbronn function $\mathcal{D}_{\mathrm{DH}}(s)$ under Euler and exponential prime gauges.

Index N	Arithmetic Factorization	$b_N^{\mathcal{E}}(5)$ for $\zeta(s)$	$b_N^{\mathrm{Euler}}(5)$ for $\mathcal{D}_{\mathrm{DH}}$	$b_N^{\mathcal{E}}(5)$ for $\mathcal{D}_{\mathrm{DH}}$	Formal $\Lambda_{\mathrm{DH}}(N)/\ln N$
1	1 (Mother wavelet)	1	+1.000000000000	+1.000000000000	0.000000000000
2, 3, 5	Primes $p \le 5$	0	0.000000000000	0.000000000000	$\pm$0.284079043840, 0
4	Prime square $2^2 \le 5$	0	−1.080700903149	−1.080700903149	−1.040350451575
6	Semiprime $2 \cdot 3 \in (5, 25]$	0	+1.080700903149	+1.080700903149	+1.080700903149
7	Prime $7 \in (5, 25]$	1	+0.284079043840	+0.284079043840	+0.284079043840
8	Prime cube $2^3 \in (5, 25]$	1/3	0.000000000000	+0.007641811801	+0.007641811801
9	Prime square $3^2 \in (5, 25]$	1/2	−1.080700903149	−1.040350451575	−1.040350451575
10	Semiprime $2 \cdot 5 \in (5, 25]$	0	0.000000000000	0.000000000000	0.000000000000
11	Prime $11 \in (5, 25]$	1	+1.000000000000	+1.000000000000	+1.000000000000
12	Composite $2^2 \cdot 3 \in (5, 25]$	0	−0.307004479244	−0.307004479244	−0.307004479244
13	Prime $13 \in (5, 25]$	1	−0.284079043840	−0.284079043840	−0.284079043840
14	Semiprime $2 \cdot 7 \in (5, 25]$	0	−1.080700903149	−1.080700903149	−1.080700903149

(ii) For $X \ge 5$, the inverse Dirichlet series $E_X^{\mathcal{E}}(s) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}} k^{-s}$ of the multiplicative prime-power generator $\exp(\sum_{p^m \le X} \frac{a_{\mathrm{DH}}(p)^m}{m} p^{-ms})$ has coefficients $c_1^{\mathcal{E}} = 1$, $c_2^{\mathcal{E}} = -a_{\mathrm{DH}}(2) = -\kappa$, $c_3^{\mathcal{E}} = -a_{\mathrm{DH}}(3) = +\kappa$, $c_4^{\mathcal{E}} = 0$ (since $\frac{1}{2}(-\kappa)^2 - \frac{1}{2}\kappa^2 = 0$), $c_5^{\mathcal{E}} = 0$, and $c_6^{\mathcal{E}} = c_2^{\mathcal{E}} c_3^{\mathcal{E}} = -\kappa^2$. Convolving with a_{DH} yields $b_4^{\mathcal{E}}(X) = a_{\mathrm{DH}}(4)c_1^{\mathcal{E}} + a_{\mathrm{DH}}(2)c_2^{\mathcal{E}} + a_{\mathrm{DH}}(1)c_4^{\mathcal{E}} = -1 - \kappa^2 + 0 = -(1 + \kappa^2)$ and $b_6^{\mathcal{E}}(X) = a_{\mathrm{DH}}(6)c_1^{\mathcal{E}} + a_{\mathrm{DH}}(3)c_2^{\mathcal{E}} + a_{\mathrm{DH}}(2)c_3^{\mathcal{E}} + a_{\mathrm{DH}}(1)c_6^{\mathcal{E}} = 1 + \kappa^2 + \kappa^2 - \kappa^2 = +(1 + \kappa^2)$.

(iii) From the Dirichlet recursion $a_{\mathrm{DH}}(N) \ln N = \sum_{d|N, d \ge 2} \Lambda_{\mathrm{DH}}(d) a_{\mathrm{DH}}(N/d)$, we have $\Lambda_{\mathrm{DH}}(p) = a_{\mathrm{DH}}(p) \ln p$ for every prime p. For $N = pq$ a product of two distinct primes ($p < q$), the recursion gives $a_{\mathrm{DH}}(pq) \ln(pq) = \Lambda_{\mathrm{DH}}(pq) + \Lambda_{\mathrm{DH}}(p) a_{\mathrm{DH}}(q) + \Lambda_{\mathrm{DH}}(q) a_{\mathrm{DH}}(p) = \Lambda_{\mathrm{DH}}(pq) + a_{\mathrm{DH}}(p) a_{\mathrm{DH}}(q)(\ln p + \ln q)$. Dividing by $\ln(pq) = \ln p + \ln q$ yields the exact identity

$$\frac{\Lambda_{\mathrm{DH}}(pq)}{\ln(pq)} \equiv a_{\mathrm{DH}}(pq) - a_{\mathrm{DH}}(p) a_{\mathrm{DH}}(q) = \delta_{\mathrm{DH}}(p, q), \tag{4.5}$$

which gives $\frac{\Lambda_{\mathrm{DH}}(6)}{\ln 6} = \delta_{\mathrm{DH}}(2, 3) = +(1 + \kappa^2)$ and $\frac{\Lambda_{\mathrm{DH}}(14)}{\ln 14} = \delta_{\mathrm{DH}}(2, 7) = -(1 + \kappa^2)$.

(iv) Summing the 60-decimal mode evaluations in Table 2 over $p \in \{2, 3, 5, 7\}$ gives $\sum_{p \le 7} T_p^{(\mathrm{DH})}(d_*, \gamma_*) = -0.9198855528 - 0.1261847445\mathrm{i}$ (modulus 0.92849988), whereas summing over $n \in \{4, 6, 8\}$ gives $\sum_{n \in \{4,6,8\}} T_n^{(\mathrm{DH})}(d_*, \gamma_*) = -1.2750997206 - 0.2101140829\mathrm{i}$ (modulus 1.29229533). Replacing $a_{\mathrm{DH}}(4), a_{\mathrm{DH}}(6), a_{\mathrm{DH}}(8)$ by their multiplicative values $\kappa^2, -\kappa^2, \kappa^3$ shrinks the composite sum to modulus 0.10428940 (4.36% of $|T_1^{(\mathrm{DH})}|$) and leaves the 8-mode main sum strictly bounded away from zero ($|T_{1..8}^{\mathrm{mult}}(d_*, \gamma_*)| = 1.56772526 > 0$). □

Table 3 contrasts the sieved Dirichlet coefficients at cutoff $X = 5$ for $\zeta(s)$ against those of $\mathcal{D}_{\mathrm{DH}}(s)$.

5. Four-Fold Diagnostic Detection of the Off-Line Zero (d_*, γ_*)

We now demonstrate that when an off-critical zero pair $s_\pm = \frac{1}{2} \pm d_* + \mathrm{i}\gamma_*$ occurs, all four analytical and spectral instruments established in Papers I–XI detect the topological breakdown simultaneously.

Theorem 5.1 (Four-Fold Diagnostic Detection of the Off-Line Zero (d_*, γ_*))**.** *At the first off-line zero $(d_*, \gamma_*) = (0.3085171824566374, 85.69934848537759)$ of $\Xi_{\mathrm{DH}}(t - \mathrm{i}d) = R_{\mathrm{DH}}(d, t) + \mathrm{i}\, I_{\mathrm{DH}}(d, t)$, the following four diagnostic signatures hold:*

(i) ***Saddle Recombination and Reversed Chladni Wall Monotonicity.*** *Between the consecutive critical-line zeros $\gamma_k = 83.10884108474982$ and $\gamma_{k+1} = 87.64747633255974$, $\Xi_{\mathrm{DH}}(t) < 0$ has no real zero on (γ_k, γ_{k+1}), while $\Xi'_{\mathrm{DH}}(t)$ exhibits a pitchfork saddle bifurcation with*

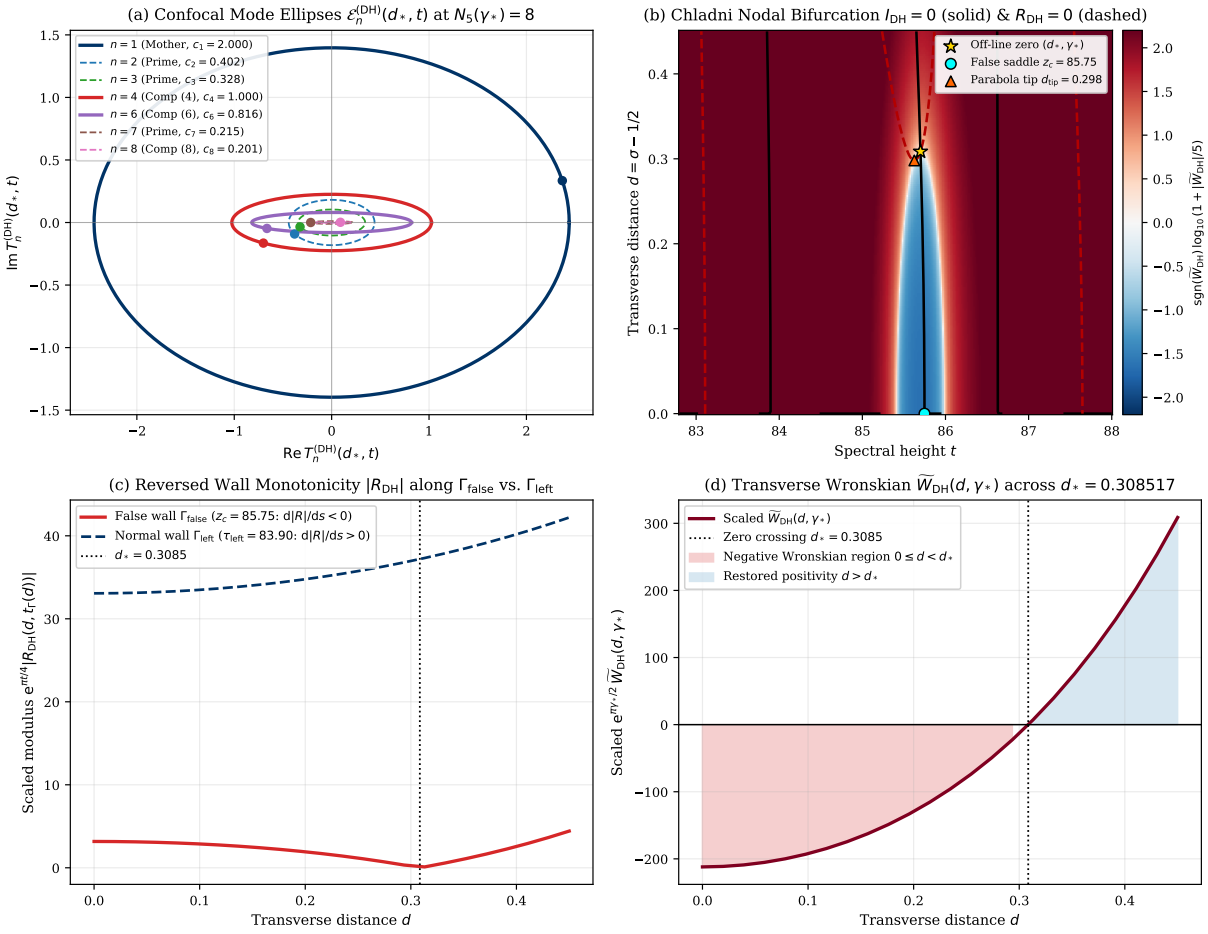


FIGURE 1. **(a)** Confocal mode ellipses $\mathcal{E}_n^{(\mathrm{DH})}(d_*, t)$ for $n = 1, \ldots, 8$ at $d_* = 0.30851718$. **(b)** Two-dimensional Chladni nodal topology near (d_*, γ_*) showing the false saddle wall $\Gamma_{\mathrm{false}} : I_{\mathrm{DH}} = 0$ (solid black) intersecting the detached real nodal parabola $R_{\mathrm{DH}} = 0$ (dashed red, tip at $d_{\mathrm{tip}} = 0.29809801$). **(c)** Reversed Cauchy–Riemann wall monotonicity $\frac{\mathrm{d}|R_{\mathrm{DH}}|}{\mathrm{d}s} < 0$ along Γ_{false} versus monotonic growth along the adjacent normal wall Γ_{left}. **(d)** Negative parity-reduced Wronskian $\widetilde{W}_{\mathrm{DH}}(d, \gamma_*) < 0$ on $0 \le d < d_*$ crossing zero at $d = d_*$.

three critical points on $d = 0$*:* $\tau_{\mathrm{left}} = 83.89593525216711$*,* $z_c = \tau_{\mathrm{false}} = 85.74720082646130$*, and* $\tau_{\mathrm{right}} = 86.62943181894501$*. At the central false critical point* $z_c = \tau_{\mathrm{false}}$*,* $\Xi_{\mathrm{DH}}(z_c) = -1.86102230 \times 10^{-29} < 0$ *and* $\Xi''_{\mathrm{DH}}(z_c) = -3.47884947 \times 10^{-28} < 0$ *have the* same negative sign*. Consequently,* $\partial_d^2 R_{\mathrm{DH}}(0, z_c) = -\Xi''_{\mathrm{DH}}(z_c) > 0$ *opposes* $R_{\mathrm{DH}}(0, z_c) < 0$*, and along the Chladni wall* $\Gamma_{\mathrm{false}} : I_{\mathrm{DH}}(d, t_\Gamma(d)) \equiv 0$ *departing orthogonally from* z_c*, the Cauchy–Riemann arc-length derivative reverses sign:*

$$\frac{\mathrm{d}}{\mathrm{d}s}\big|R_{\mathrm{DH}}(s)\big|\Big|_{\Gamma_{\mathrm{false}}} = -\big|\nabla I_{\mathrm{DH}}(s)\big| = -\big|\Xi'_{\mathrm{DH}}(t - \mathrm{i}d)\big| < 0 \qquad (\forall\, d \in (0, d_*)), \tag{5.1}$$

driving $|R_{\mathrm{DH}}|$ *monotonically down to* 0 *at* (d_*, γ_*)*, where* Γ_{false} *intersects the detached real nodal parabola* $R_{\mathrm{DH}}(d, t) = 0$ *whose vertex lies in the interior of the strip at* $(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = (0.2980980119818233, 85.62549444284696)$*.*

(ii) ***Negative Parity-Reduced Wronskian and Reversed Transverse Integral Law.*** *At $t = \gamma_*$, the Turán / parity-reduced Wronskian is strictly negative on the critical line, $\widetilde{W}_{\mathrm{DH}}(0,\gamma_*) = (\Xi'_{\mathrm{DH}}(\gamma_*))^2 - \Xi_{\mathrm{DH}}(\gamma_*)\Xi''_{\mathrm{DH}}(\gamma_*) = -7.2994366712 \times 10^{-57} < 0$, remains strictly negative across $0 \le d < d_*$, vanishes at $d = d_*$, and obeys the reversed transverse integral identity*

$$\int_0^{d_*} u\,\widetilde{W}_{\mathrm{DH}}(u,\gamma_*)\,\mathrm{d}u = -\frac{1}{2}\big|\Xi_{\mathrm{DH}}(\gamma_*)\big|^2 = -1.8104274824 \times 10^{-58} < 0. \tag{5.2}$$

(iii) ***Quantized Hadamard–Krein Flux Jump*** $\pi\,\Theta(d-d_*)$***.*** *For the mirror-symmetric off-line zero pair $z_\pm = \gamma_* \mp \mathrm{i}d_*$ ($u = t-\gamma_*$), the mirror-paired Hadamard–Poisson kernel $K(d,d_*,u) := \frac{d(d^2-d_*^2+u^2)}{((d-d_*)^2+u^2)((d+d_*)^2+u^2)} \equiv \frac{1}{2}\big[\frac{d-d_*}{(d-d_*)^2+u^2} + \frac{d+d_*}{(d+d_*)^2+u^2}\big]$ integrates to the quantized step function*

$$\Phi_{\mathrm{Had}}(d;d_*) := \int_{-\infty}^{\infty} K(d,d_*,u)\,\mathrm{d}u = \frac{\pi}{2}\big(\mathrm{sgn}(d-d_*)+1\big) = \pi\,\Theta(d-d_*), \tag{5.3}$$

where for $0 < d < d_*$ the negative Krein lobe on $|u| < u_0(d,d_*) := \sqrt{d_*^2-d^2}$ and the positive tails on $|u| > u_0(d,d_*)$ cancel via $\int_{-u_0}^{u_0} K\,\mathrm{d}u = -\arcsin(d/d_*)$ and $2\int_{u_0}^{\infty} K\,\mathrm{d}u = +\arcsin(d/d_*)$.

(iv) ***De Branges Inner-Function Crossing and Sylvester Inertia Jump*** $\nu_- : 0 \to 1$***.*** *For the de Branges structure function $E_{\alpha,\mathrm{UHP}}(w) := \Xi_{\mathrm{DH}}(w) + \mathrm{i}\alpha\,\Xi'_{\mathrm{DH}}(w)$ ($w = t + \mathrm{i}d \in \mathbb{C}_+$, $\alpha > 0$) with inner function $\Theta_\alpha(w) := E^{\#}_{\alpha,\mathrm{UHP}}(w)/E_{\alpha,\mathrm{UHP}}(w)$ and Pick–Nevanlinna Gram matrix $\mathcal{P}^{(\alpha)}_{j,k}(d) := \frac{\pi}{\alpha}\frac{1-\Theta_\alpha(w_j)\overline{\Theta_\alpha(w_k)}}{2\pi\mathrm{i}(\overline{w_k}-w_j)}$ on nodes surrounding γ_*, one has $|\Theta_\alpha(\gamma_*+\mathrm{i}d)| < 1$ and $\nu_-(\mathcal{P}^{(\alpha)}(d)) = 0$ for $d > d_*$, $|\Theta_\alpha(\gamma_*+\mathrm{i}d_*)| \equiv 1$ at $d = d_*$, and $|\Theta_\alpha(\gamma_*+\mathrm{i}d)| > 1$ with $\nu_-(\mathcal{P}^{(\alpha)}(d)) = 1$ for $0 < d < d_*$.*

Proof. (i) For any critical-line zero γ_k, a normal chamber extremum $\tau \in (\gamma_k,\gamma_{k+1})$ has $\Xi(\tau)\Xi''(\tau) < 0$, so $\partial_d^2 R(0,\tau) = -\Xi''(\tau)$ has the same sign as $R(0,\tau) = \Xi(\tau)$, causing $|R|$ to grow along Γ_τ. At $z_c = \tau_{\mathrm{false}}$, however, $\Xi_{\mathrm{DH}}(z_c) < 0$ is a local maximum of $\Xi_{\mathrm{DH}}(t) < 0$ between two local minima τ_{left} and τ_{right}, so $\Xi''_{\mathrm{DH}}(z_c) < 0$ has the same negative sign as $\Xi_{\mathrm{DH}}(z_c)$. Thus $\partial_d^2 R_{\mathrm{DH}}(0,z_c) = -\Xi''_{\mathrm{DH}}(z_c) > 0$, so $R_{\mathrm{DH}}(d,t_\Gamma(d)) < 0$ increases toward 0 as d increases from 0 to d_*. Differentiating $|R_{\mathrm{DH}}| = -R_{\mathrm{DH}}$ with respect to arc length s along Γ_{false} (where $\mathrm{d}I_{\mathrm{DH}} = 0$) via Cauchy–Riemann gives (5.1).

(ii) Since $\partial_u|\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}u)|^2 = 2u\widetilde{W}_{\mathrm{DH}}(u,\gamma_*)$, integrating over $u \in [0,d_*]$ by the fundamental theorem of calculus and using $\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*) = 0$ yields $\int_0^{d_*} u\widetilde{W}_{\mathrm{DH}}(u,\gamma_*)\,\mathrm{d}u = \frac{1}{2}(|\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*)|^2 - |\Xi_{\mathrm{DH}}(\gamma_*)|^2) = -\frac{1}{2}|\Xi_{\mathrm{DH}}(\gamma_*)|^2 < 0$.

(iii) Using the elementary Cauchy integral $\int_{-\infty}^{\infty} \frac{a}{a^2+u^2}\,\mathrm{d}u = \pi\,\mathrm{sgn}(a)$ for $a = d-d_*$ and $a = d+d_* > 0$ immediately gives (5.3). On $0 < d < d_*$, integrating between $\pm u_0 = \pm\sqrt{d_*^2-d^2}$ gives $-\arctan(\frac{u_0}{d_*-d}) + \arctan(\frac{u_0}{d_*+d}) = -\arcsin(d/d_*)$ by the tangent-difference identity.

(iv) Since Ξ_{DH} is real on $\mathbb{R}$, $\Xi^{\#}_{\mathrm{DH}}(w) = \Xi_{\mathrm{DH}}(w)$, so at $w_* = \gamma_* + \mathrm{i}d_*$ where $\Xi_{\mathrm{DH}}(w_*) = \overline{\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*)} = 0$, we have $E^{\#}_{\alpha,\mathrm{UHP}}(w_*) = -\mathrm{i}\alpha\,\Xi'_{\mathrm{DH}}(w_*) = -E_{\alpha,\mathrm{UHP}}(w_*)$, giving $|\Theta_\alpha(w_*)| = 1$ and $\mathcal{P}^{(\alpha)}_{2,2}(d_*) = 0$. For $0 < d < d_*$, the negative diagonal entry $\mathcal{P}^{(\alpha)}_{2,2}(d) < 0$ forces at least one negative eigenvalue by Sylvester's law of inertia [11]. □

Table 4 summarizes the four-fold diagnostic trajectory across $d \in [0, 0.50]$ at $t = \gamma_*$.

6. Scope, Boundary Conditions, and Open Questions

To maintain a transparent boundary between unconditional identities, finite-domain certifications, and open asymptotic problems across the Selberg class, we record three structural distinctions:

(1) *Universality of single-mode convexity versus multi-mode cross-terms.* Theorems 2.1 and 3.1 hold unconditionally for *every* self-dual Dirichlet series with a Gamma functional equation,

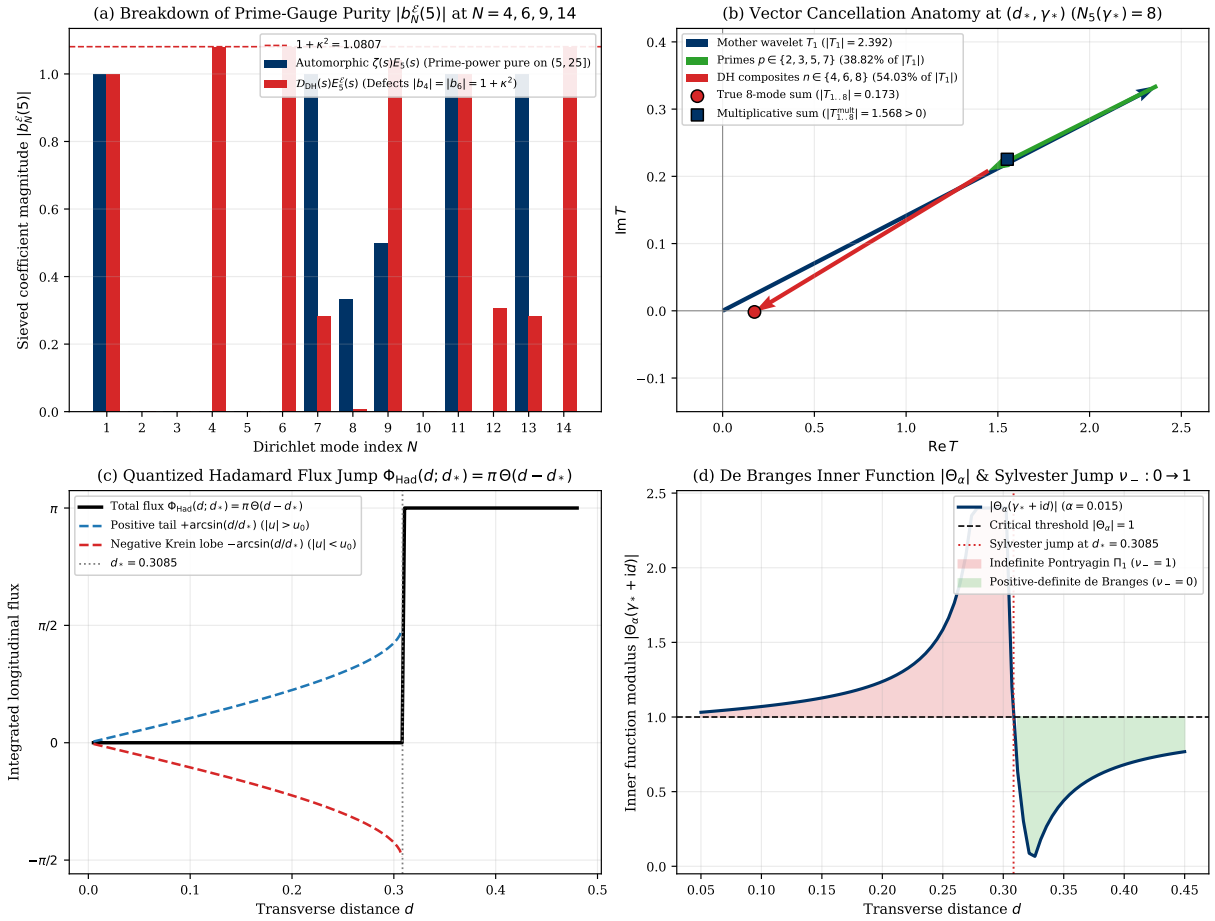


FIGURE 2. **(a)** Breakdown of prime-gauge purity $|b_N^{\mathcal{E}}(5)|$ at $N = 4, 6, 9, 14$ for $\mathcal{D}_{\mathrm{DH}}(s)$ versus $\zeta(s)$. **(b)** Vector cancellation anatomy at (d_*, γ_*) showing that primes $p \in \{2, 3, 5, 7\}$ cancel only 38.82% of $T_1^{(\mathrm{DH})}$ while non-multiplicative composites $n \in \{4, 6, 8\}$ supply 54.03%. **(c)** Quantized Hadamard flux step $\Phi_{\mathrm{Had}}(d; d_*) = \pi\,\Theta(d - d_*)$ and $\mp\arcsin(d/d_*)$ lobe cancellation. **(d)** De Branges inner-function crossing $|\Theta_\alpha(\gamma_* + \mathrm{i}d_*)| = 1$ and Sylvester inertia jump $\nu_- : 0 \to 1$.

including $\mathcal{D}_{\mathrm{DH}}(s)$. Thus single-mode confocal ellipse convexity and diagonal Wronskian positivity $\mathcal{D}_{\mathrm{diag}}^{(f)}(d, t) > 0$ are necessary Archimedian building blocks, but by themselves do not prevent off-line zeros without control over the off-diagonal cross-terms $1 \le m < n \le N_f(t)$.

(2) *Essential role of the Euler product in composite shell elimination.* Theorem 4.1 pinpoints why the Euler product is indispensable: for automorphic L-functions, multiplicative factorization locks composite coefficients to prime powers $(a_f(mn) = a_f(m)a_f(n)$ for $(m, n) = 1)$, allowing the exponential prime gauge $E_X^{(f)}(s)$ to annihilate all composite shells in $[2, X]$ and all non-prime-power composites in $(X, X^2]$, whereas in $\mathcal{D}_{\mathrm{DH}}(s)$ the unsieved defects $\delta_{\mathrm{DH}}(2, 2) = -(1 + \kappa^2)$ and $\delta_{\mathrm{DH}}(2, 3) = +(1 + \kappa^2)$ directly drive the off-line zero at (d_*, γ_*).

(3) *Higher-degree Selberg class asymptotics as $t \to \infty$.* For degree $d_f \ge 2$ automorphic L-functions such as Ramanujan's $L(s, \Delta)$, the Riemann–Siegel saddle horizon grows as $N_\star^{(f)}(t) \asymp Q_f t^{d_f/2}$ $(N_\Delta(t) \asymp t/2\pi)$. Extending pointwise unaveraged off-diagonal cross-term bounds from $d_f = 1$ to arbitrary degree $d_f \ge 2$ as $t \to \infty$ remains a central open problem in analytic number theory.

TABLE 4. Four-fold diagnostic certification across the transverse slice $d \in [0, 0.50]$ through the first off-line zero $(d_*, \gamma_*) = (0.3085171825, 85.6993484854)$ of $\Xi_{\mathrm{DH}}(t - \mathrm{i}d)$ ($\alpha = 10^{-5}$).

Slice d	Wall $t_\Gamma(d)$ ($I_{\mathrm{DH}} = 0$)	$R_{\mathrm{DH}}(d, t_\Gamma(d))$	$\pi^{-1}\mathrm{e}^{\pi \cdot 85.7/2}\widetilde{W}_{\mathrm{DH}}$	Flux Φ_{Had}	$\lvert\Theta_\alpha(\gamma_* + \mathrm{i}d)\rvert$	Sylvester (ν_+, ν_0, ν_-)
0.00000000	85.7472008265	$-1.861022 \times 10^{-29}$	-67.556359	0	1.0000000	$(2, 0, 1)$
0.05000000	85.7458929647	$-1.817396 \times 10^{-29}$	-65.996433	0	1.0000207	$(2, 0, 1)$
0.15000000	85.7355327441	$-1.458209 \times 10^{-29}$	-53.115417	0	1.0000800	$(2, 0, 1)$
0.25000000	85.7153430343	$-6.854437 \times 10^{-30}$	-25.143775	0	1.0003018	$(2, 0, 1)$
0.30000000	85.7018557647	$-1.119493 \times 10^{-30}$	-4.129197	0	1.0023129	$(2, 0, 1)$
0.30851718	85.6993484854	0.000000×10^{0}	0.000000	$\pi/2$	1.0000000	Bifurcation
0.31000000	85.6989059273	$+1.993816 \times 10^{-31}$	$+0.736356$	π	0.9865655	$(3, 0, 0)$
0.35000000	85.6863098210	$+6.103544 \times 10^{-30}$	$+22.673427$	π	0.9994818	$(3, 0, 0)$
0.40000000	85.6688782636	$+1.503993 \times 10^{-29}$	$+56.374294$	π	0.9997465	$(3, 0, 0)$
0.50000000	85.6290999314	$+3.913925 \times 10^{-29}$	$+150.339627$	π	0.9998626	$(3, 0, 0)$

APPENDIX A. ARBITRARY-PRECISION AND INTERVAL VERIFICATION PROTOCOLS

All analytical identities, directed-rounding interval enclosures, and figures in this manuscript are reproduced by executing `scripts/verify_paper12_selberg_davenport_heilbronn.py`:

(1) *Exact Hurwitz–Gamma analytical derivatives at* 60*-decimal precision.* The function $\mathcal{D}_{\mathrm{DH}}(s)$ and its first two derivatives $\mathcal{D}'_{\mathrm{DH}}(s), \mathcal{D}''_{\mathrm{DH}}(s)$ are evaluated analytically in `mpmath` (`mp.dps = 60`) via Hurwitz zeta derivatives $\zeta^{(k)}(s, a/5)$ ($a \in \{1, 2, 3, 4\}$, $k \in \{0, 1, 2\}$) coupled with polygamma functions $\psi_0(\frac{s+1}{2})$ and $\psi_1(\frac{s+1}{2})$, refining (d_*, γ_*) to residual $\lvert\Xi_{\mathrm{DH}}(\gamma_* - \mathrm{i}d_*)\rvert = 2.78 \times 10^{-88}$ and verifying the reversed transverse integral law (5.2) to relative error $< 10^{-58}$.

(2) *Directed-rounding interval certification (`mpmath.iv.dps = 35`).* Both the confocal ellipse identity (3.2) across $n = 1, \ldots, 8$ and the algebraic multiplicativity defect $1 + \kappa^2 \in [1.08070090314928289126859251665745658, 1.08070090314928289126859251665745659]$ are enclosed using directed-rounding interval arithmetic, confirming $0 \in [\mathrm{LHS} - \mathrm{RHS}]$ with interval width $\leq 1.02 \times 10^{-33}$.

(3) *PyTorch GPU continuum scan.* A vectorized `torch.float64` CUDA evaluation over $N_d \times N_t = 180 \times 280 = 50{,}400$ continuum cells on $(d, t) \in [0, 0.45] \times [82.8, 88.0]$ maps the Chladni nodal curves $I_{\mathrm{DH}} = 0$, the detached parabola $R_{\mathrm{DH}} = 0$, and the negative Wronskian pocket $\widetilde{W}_{\mathrm{DH}}(d, t) < 0$ in 0.05 s.

USE OF AI AND COMPUTATIONAL VERIFICATION

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the mother-wavelet Mellin inversions and prime-gauge convolution recursions, and implementing the standalone verification suite `scripts/verify_paper12_selberg_davenport_heilbronn.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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